

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q13-0192

Fossil 100 kV  
January 01, 2015

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0192. The Interconnection Customer proposes a 192.8 MW natural gas combined cycle facility interconnecting at the Fossil 100 kV substation in the Michigan area. The request was received on January 02, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 122 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,277,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                          | Contingency                             | Loading (%) | Rating (MVA) | DFAX (%) |
|---------------------------------------------|-----------------------------------------|-------------|--------------|----------|
| IMPERIAL VLY – S CLEVELAND 100 kV           | OCEANSIDE – PALISADES SUBSTATION 500 kV | 104.9       | 130          | 10.8     |
| S CLEVELAND – CHATTANOOGA PSCC 100 kV L     | OCEANSIDE – PALISADES 500 kV Line       | 102.7       | 126          | 10.8     |
| ALMENA SUBSTATION – CHATTANOOGA PSCC 100 kV | PALISADES – OCEANSIDE 500 kV            | 100.4       | 121          | 10.8     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                    | Required Upgrade           | Cost Estimate | Allocation to Project |
|---------------------------------------------|----------------------------|---------------|-----------------------|
| SOUTH CLEVELAND – IMPERIAL VLY 100 kV line  | Reconductor line           | \$4,291,000   | 100% (\$4,291,000)    |
| CHATTANOOGA PSCC – SOUTH CLEVELAND 100 kV L | Reconductor line           | \$3,650,000   | 100% (\$3,650,000)    |
| ALMENA SUBSTATION – CHATTANOOGA PSCC 100 kV | Reconductor line           | \$2,336,000   | 100% (\$2,336,000)    |
|                                             | Total allocated to project |               | \$10,277,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.